Mathematics for Economics

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课程简介

- 按照“授人以渔”的理念，我们会从topology的基础开始
  – “C” words: convergence, continuity, connectedness, completeness, compactness, and category
- 或者会涉猎functional analysis, measure theory
- Optimization?
  – 无限维的Lagrangian method
  – Theorem of Maximum?
  – ...

课程简介

参考读物包括:

- *Topology*, Munkres
- *Real Analysis*, Carothers
- *Optimization by Vector Space Methods*, Luenberger

Set Theory and Logic

- object/element a is in set A: $a \in A$
- not in: $\notin$
- logical identity: $=$
- subset/inclusion: $\subset$ or $\subseteq$
- proper subset/inclusion: $\subsetneq$
- natural language $\leftrightarrow$ math symbol

Set Theory and Logic

To build a set from scratch:

- list all the elements, $A = \{a, b, c\}$
- or describe the property possessed by the elements:
  – B is the set of all x such that x is an even integer:
  $B = \{x \mid x \text{ is an even integer}\}$

To build a set from previously defined sets:

- union: $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$
  – the statement "P or Q" means "P or Q, or both"
- intersection: $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$
- empty set: $\varnothing$
- difference/complement: $A - B = \{x \mid x \in A \text{ and } x \notin B\}$

Set Theory and Logic

To build a "new" statement from "old" statements...

- P or Q: $P \vee Q$
- P and Q: $P \wedge Q$
- if P (hypothesis), then Q (conclusion): $P \implies Q$
  - if P is true, then Q is true
  - if P is false, then Q may be either true of false
- negation of P, i.e. P is not true: $\neg P$
- contrapositive: $\neg Q \implies \neg P$
- converse: $Q \implies P$
  - Q is true, only if P is true

Truth value... $P \implies Q = (\neg P) \vee Q$?

Set Theory and Logic

- Logical quantifiers:
  - for each/for every: $\forall$
  - there exists/for at least one: $\exists$
  - how do they relate to "or" and "and"?
- Negation of quantifies:
  - $\neg\left(\forall x \in A, P\right) = \left(\exists x \in A, \neg P\right)$
  - $\neg\left(\exists x \in A, P\right) = \left(\forall x \in A, \neg P\right)$

Set Theory and Logic

Continue to build new sets

- collection/family/set of sets:
  - $\mathcal{A} = \{A, B, C...\}$
  - $\mathcal{P}(A) = \{A' \mid A' \subseteq A\}$, $\mathcal{P}(\mathcal{P}(A))$, ...
- arbitrary union/intersection:
  - $\bigcup_{A \in \mathcal{A}} A = \{x \mid \exists A \in \mathcal{A}, x \in A\}$
  - $\bigcap_{A \in \mathcal{A}} A = \{x \mid \forall A \in \mathcal{A}, x \in A\}$
- Cartesian product: $A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$

Set Theory and Logic

Rules of set theory

- $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$...
- $A - (B \cap C) = (A - B) \cup (A - C)$...
- how to prove set identity?
  – $A = B$: $x \in A \iff x \in B$, "$A \subseteq B$ and $B \subseteq A$"

Set Theory and Logic

Relating set theory and logic:

- set and statement; empty and false
- union/intersection and or/and
- difference/complement and negation
- subset and "if...then..."
- category theory?

One of union, intersection and complement is redundant/derived?

Set Theory and Logic

What we have done?

- Define from scratch some math objects
- Construct new objects through operations on old objects
- Relate different categories of objects

Functions

To define a function/mapping from scratch:

- a function f is a subset of $X \times Y$ such that:
  – $(x, y) \in f$ and $(x, y') \in f$ implies $y = y'$
- domain of f: $A = \{x \in X \mid \exists y \in Y, (x, y) \in f\}$
- image of f: $B = \{y \in Y \mid \exists x \in X, (x, y) \in f\}$
- we can write $f : A \to Y$ with Y understood as the range of f, but we cannot write $f : X \to Y$ if $\mathrm{dom}(f)$ does not contain X
- what is a well-defined function?
  – or we ask, what does it mean by ill-defined?

Functions

To define new functions from old functions:

- restriction: given $f : A \to B$ and $A_0 \subset A$, the restriction of f to A_0 is $f|_{A_0} : A_0 \to B$ defined by $(f|_{A_0})(a_0) = f(a_0)$
- composition: given $f : A \to B$ and $g : B \to C$, the composite of f and g is $g \circ f : A \to C$ defined by $(g \circ f)(a) = g(f(a))$
- how about summation, product, and etc.?
  – we haven't defined any operation on B yet
  – in fact, "operation" is a kind of function

Functions

Some definitions:

- $f : A \to Y$ is injective (or is an injection) if $a_1 \neq a_2 \Longrightarrow f(a_1) \neq f(a_2)$
- $f : A \to Y$ is surjective (or is an surjection) if $\text{Image}(f) = f(A) = Y$
- $f : A \to Y$ is bijective (or is an bijection) if it is both injective and surjective

Functions

To define new functions from old functions:

- inverse: the inverse of a bijection $f : A \to B$ is $f^{-1} : B \to A$ defined by $f(f^{-1}(b)) = b$
  – why is f^{-1} well-defined?
- image of $A_0 \subset A$ and preimage of $B_0 \subset B$
  – the image $f(A_0) = \{b \in B \mid \exists a \in A_0, b = f(a)\}$
  – even if f is neither injective nor surjective, we can define the preimage $f^{-1}(B_0) = \{a \in A \mid f(a) \in B_0\}$

Functions

Exercise: Let $f : A \to B$. If there are functions $g : B \to A$ and $h : B \to A$ such that $\forall a \in A, g(f(a)) = a$, and $\forall b \in B, f(h(b)) = b$, then f is bijective and $g = h = f^{-1}$.

Relations

- We have defined a function as a subset of a cartesian product.
- Intuitively, a and b can be *related* by $f(a) = b$
- One a can be related to only one b
- What if we want to talk about "relation" in general?
- As economists, we care about preference, correlation/causality and etc., which are all relations

Relations

- A relation on A is a subset of $A \times A$
  - x is in the relation R to y: $(x, y) \in R$ or xRy
  - A function is a special relation
- Equivalence and Order are two relations of interest for mathematicians and economists
  - indifference is an equivalence relation
  - we talk about outcome/ payoff/strategy equivalence
  - strict preference is an order relation
  - we also talk about monotone strategy/equilibrium

Relations

An equivalence relation on A is $I \subset A \times A$ such that:

- (reflexivity) $\forall x \in A, xIx$
- (symmetry) $xIy \implies yIx$
- (transitivity) If xIy and yIz, then xIz

We often use "$\sim$" to denote an equivalence relation.

Relations

Given an equivalence relation $\sim$ and $x \in A$, an **equivalence class** determined by x is $E = \{y \mid y \sim x\}$
Note that $x \in E$ (why?)

Lemma

If E and E' are equivalent classes, then either $E \cap E' = \varnothing$ (disjoint) or $E = E'$ (equal).

Proof.

Let E be determined by x and E' be determined by x'. Suppose $E \cap E' \neq \varnothing$, i.e. there exists $y \in E \cap E'$, we want to show that $E = E'$.
By definition, $y \sim x$ and $y \sim x'$. By symmetry, $x \sim y$ and $y \sim x'$. By transitivity, $x \sim x'$. For each $z \in E$, we have $z \sim x \sim x'$, and thus $z \in E'$ and $E \subseteq E'$. Similarly, we can show that $E' \subseteq E$... $\square$

Relations

- Note that an indifference curve is an equivalence class, and "different indifference curves do not intersect" is simply restating the lemma in a special case.
- A partition of A is a collection of disjoint nonempty subsets of A whose union is all of A
- An equivalence relation on A defines a *unique* partition on A
- A partition defines a *unique* equivalence relation
  – how to show uniqueness?

Relations

An (simple) order relation on A is $P \subset A \times A$ such that:

- (comparability) $\forall x, y \in A$ with $x \neq y$, either xPy or yPx
- (non-reflexivity) $\forall x \in A$, $(x, x) \notin P$
- (transitivity) ...

I prefer to use "$\prec$"(precede) to denote an order relation and leave "$<$"(less than) for the real line; $\preceq$ denotes "$\prec$ or identical"

Economists may need to work with relations that fails comparability, e.g. Pareto

Relations

Exercise: Here is a "proof" that every relation C that is both symmetric and transitive is also reflexive: "Since C is symmetric, aCb implies bCa. Since C is transitive, aCb and bCa imply aCa, as desired." Find the flaw in this argument.

Relations

A relation provides a structure on a set:

- an equivalence partitions a set
- an order "linearizes" a set
  – the term "linear" is not super accurate; lexicographic order

And we can compare different structured sets

- $f : (A, \prec_A) \to (B, \prec_B)$ is an order-preserving bijection if

$$a_1 \prec_A a_2 \implies f(a_1) \prec_B f(a_2)$$

  – why not "$\iff$"?
  – how about "relation-preserving"?
- $(A, \prec_A)$ and $(B, \prec_B)$ are of the same order type if there exists an order-preserving bijection between them
- Example 10 on Pg. 25

Relations

Consider $(A, \prec)$ and $A_0 \subset A$

- b is the largest element of A_0 if $b \in A_0$ and if $\forall x \in A_0$, $x \preceq b$
- smallest element...
- b is the maximal/undominated element of A_0 if there is no $a \in A_0$ s.t. $b \prec a_0$
  – when do "maximal" and "undominated" differ?
- A_0 is bounded above if there is $b \in A$ s.t. $\forall x \in A_0, x \preceq b$
- b is called an upper bound for A_0

Relations

- the least upper bound/supremum of A_0, if well-defined, is

$$\sup A_0 = \min \{b \in A \mid \forall x \in A_0, x \preceq b\}$$

- bounded below, lower bound, greatest lower bound/infimum
- $(A, \prec)$ has the least upper bound property if each non-empty subset A_0 of A that is bounded above has a least upper bound

Excercise: $(A, \prec)$ has the least upper bound property if and only if it has the greatest lower bound property

Integers and real numbers

- Binary operation: $f : A \times A \to A$
  – e.g. $+$ (summation/plus), $\cdot$ (multiplication/product)
- $\mathbb{Z}$ and $\mathbb{R}$ can be defined by algebraic and order properties
- $\mathbb{Z}_+$ is the smallest inductive subset of $\mathbb{R}$
  – Principle of induction

Theorem

(Well-ordering) Every non-empty subset of $\mathbb{Z}_+$ has a smallest element.

Cartesian products revisited

With the notion of function, we can define cartesian products in general: let $\mathcal{A}$ be a non-empty family of sets,

- an indexing function for $\mathcal{A}$ is a surjective function $f : J \to \mathcal{A}$
  – f is not required to be injective
- J is called the index set; $\forall \alpha \in J$, write $f(\alpha) = A_\alpha$
- $\{\mathcal{A}_\alpha\}_{\alpha \in J}$ is called an indexed family of sets
- union of $\mathcal{A}$: $\bigcup_{\alpha \in J} A_\alpha = \{x \mid \exists \alpha \in J, x \in A_\alpha\}$

Cartesian products revisited

- given an index set J and a set X, a J-tuple of elements of X is a function:

$$\boldsymbol{x} : J \to X$$

  with $x_\alpha = \boldsymbol{x}(\alpha)$ denoting the α-th coordinate of $\boldsymbol{x}$
  – $J = \{1, 2, ..., m\}$ and m-tuple; X^m
  – $J = \mathbb{Z}_+$ and ω-tuple; X^ω
- the J-product of X is the set of of all J-tuples, denoted as X^J
- the Cartesian product of $\{\mathcal{A}_\alpha\}_{\alpha \in J}$, denoted as $\prod_{\alpha \in J} A_\alpha$, is the set of $\boldsymbol{x} \in \left(\bigcup_{\alpha \in J} A_\alpha\right)^J$ s.t. $x_\alpha \in A_\alpha$

Exercise: Find a bijection $f : X^\omega \times X^\omega \to X^\omega$

Finite sets

Frequently, we (will) see terms like finite, infinite, countable, uncountable, and cardinality, but how to understand them mathematically?

- Recall that $S_{n+1} = \{1, ..., n\}$ is the n-section of $\mathbb{Z}_+$
  – S_{n+1} should be a finite set
- A is finite if it is empty, or if there is a bijection $f : A \to S_{n+1}$ for some $n \in \mathbb{Z}_+$
- When A is empty, we say that A has cardinality 0
- When there is a bijection $f : A \to S_{n+1}$ we say that A has cardinality n

Finite sets

Does a finite set have a unique cardinality?

- This question looks naive, but its proof is non-trivial (see also Ex.1)
- (Lemma 6.1) Let A be a finite set and let $a_0 \in A$. Then $\text{card}(A) = n + 1$ if and only if $\text{card}(A - \{a_0\}) = n$
- (Theorem 6.2) Suppose $\text{card}(A) = n$, then for each proper subset B of A, $\text{card}(B) < n$
- If A is finite, then there is no bijection of A with a proper subset of itself
  – $\mathbb{Z}_+$ is not finite (infinite)
- $\text{card}(A)$ is indeed well-defined

Finite sets

Other properties of finite sets:

- If A is finite and $B \subset A$, then B is finite. If $B \subsetneq A$, then $\text{card}(B) < \text{card}(A)$

Corollary

Suppose $B \neq \varnothing$. Then the following are equivalent:

(1) *B is finite*
(2) *There is a surjection $f : S_{n+1} \to B$ for some n*
(3) *There is an injection $g : B \to S_{m+1}$ for some m*

- Finite unions and finite Cartesian products of finite sets are finite

Countable and uncountable sets

- A is infinite if it is not finite
- A is countably infinite if there is a bijection $f : A \to \mathbb{Z}_+$
  - $\mathbb{Z}$ is countably infinite
  - the set of even positive integers is countably infinite
  - proper inclusion need not imply different cardinalities
- A is countable if it is either finite or countably infinite; o.w. A is uncountable

Theorem

Suppose $B \neq \varnothing$. The the following are equivalent:

(1) *B is countable*

(2) *There is a surjection $f : \mathbb{Z}_+ \to B$*

(3) *There is an injection $g : B \to \mathbb{Z}_+$*

Countable and uncountable sets

When proving (3) $\Longrightarrow$ (1), we need to show the following lemma:

Lemma

If C is an infinite subset of $\mathbb{Z}_+$, then C is countably infinite.

Proof.

We define a bijection $h : \mathbb{Z}_+ \to C$ "by induction". Let

$$h(1) = \min C$$
$$h(n) = \min \{C - h(S_n)\}$$

(1) h is injective: given $m < n$, $h(n) \notin h(S_n)$ and thus $h(n) \neq h(m)$. (2) h is surjective: given $c \in C$, there is $n \in \mathbb{Z}_+$ such that $h(n) > c$; let $m = \min \{n \mid h(n) > c\}$, then $h(m-1) = c$. $\square$

Countable and uncountable sets

As Munkres points out, the construction of h is not exactly inductive

- the definition of $h(n)$ depends on the definition of h
  – see also Ex.7.9
- principle of recursive definition
- "the barber of Seville": define $h : \mathbb{Z}_+ \to \mathbb{Z}_+$ by $h(i) = 1$ if $h(i) \neq i$
  – Seville needs at least two barbers
  – the paradox "is" non-existence of fixed points

Countable and uncountable sets

Other properties of countable sets:

- A subset of a countable set is countable
- $\mathbb{Z}_+ \times \mathbb{Z}_+$ is countable
- A countable union of countable sets is countable
- A finite product of countable sets is countable
- How about a countably infinite product of countable sets?

Countable and uncountable sets

Theorem

Let $X = \{0, 1\}$. Then X^ω is uncountable.

Proof.

We want to show that there is no surjection $g : \mathbb{Z}_+ \to X^\omega$ by a diagonal argument. Write

$$g(n) = (x_{n1}, x_{n2}, ..., x_{nm}, ...).$$

Define $\boldsymbol{y} = (y_1, y_2, ..., y_n, ...) \in X^\omega$ by

$$y_n = \begin{cases} 0 & \text{if } x_{nn} = 1 \\ 1 & \text{if } x_{nn} = 0. \end{cases}$$

Then $\boldsymbol{y} \notin g(\mathbb{Z}_+)$. □

Countable and uncountable sets

- We say that $\text{card}(A) \geq (\leq)\text{card}(B)$, if there is a surjection(injection) from A to B
- X^ω is essentially $\mathcal{P}(\mathbb{Z}_+)$ (Ex.7.2), and we just show that

$$\text{card}(\mathcal{P}(\mathbb{Z}_+)) > \text{card}(\mathbb{Z}_+),$$

which can be generalized into the following theorem

Theorem

Let A be a set. There is no injection $f : \mathcal{P}(A) \to A$, and there is no surjection $g : A \to \mathcal{P}(A)$.

Too intuitive to be easy

The following are equivalent:

- Axiom of choice: Given a collection $\mathcal{A}$ of disjoint non-empty subsets, there exists a set C such that $C \subset \bigcup_{A \in \mathcal{A}} A$, and for each $A \in \mathcal{A}$, $C \cap A$ contains a single element (is singleton)
- Well-ordering theorem: For each set A, there exists an order relation on A that is a well-ordering
  – what is a well-ordering of $\mathbb{R}$?
- Zorn's lemma: Let $\prec$ be a partial order on A. If every totally ordered subset of A has an upper bound in A, then A has a maximal element
  – application: Szpilrajn extension theorem

Too intuitive to be easy

- A (weak) partial order relation R on A is $R \subset A \times A$ satisfying reflexivity, transitivity and anti-symmetry
  - anti-symmetry: xRy and yRx implies $x = y$
- A relation Q is an extension of R if $R \subset Q$

Theorem (Szpilrajn extension)

For each (weak) partial order R on X, there exists a total order Q on X which is an extension of R.

- R: the "partial" preference revealed by an individual's choice data
- Q: a "complete" preference consistent with the individual's choice data

Too intuitive to be easy

Proof:

- Consider the non-trivial case where there are $x \neq y$ such that neither $(x, y) \in R$ nor $(y, x) \in R$.
- We can extend R to a partial order containing (x, y).
- Firstly, add (x, y) to R and form $R \cup \{(x, y)\}$
  – $R \cup \{(x, y)\}$ may not be transitive
- Secondly, take the transitive closure of $R \cup \{(x, y)\}$ and form

$$R' = \mathrm{Tr}\left(R \cup \{(x, y)\}\right).$$

  – $\mathrm{Tr}(S)$ is the smallest transitive relation containing S: if $\tilde{S}$ is transitive and $S \subset \tilde{S}$, then $\mathrm{Tr}(S) \subset \tilde{S}$
  – $R' - R$ contains (a, b) such that aRx and yRb
  – Why is R' a partial order?

Too intuitive to be easy

- Let $\mathcal{A}$ be the set of partial orders on X containing R.
- Equip $\mathcal{A}$ with the partial order of set inclusion $\subset$.
- To apply Zorn's Lemma, we want to show that every totally ordered subset $\mathcal{A}'$ of $\mathcal{A}$ has an upper bound in $\mathcal{A}$
  - $\bigcup_{R \in \mathcal{A}'} R$ is the upper bound of $\mathcal{A}'$ in $\mathcal{A}$; need to show that $\bigcup_{R \in \mathcal{A}'} R$ is a partial order:
    – reflexivity, transitivity, anti-symmetry
- By Zorn's Lemma, $\mathcal{A}$ has a maximal element Q, and we want to show that Q is total.
  - Suppose Q is not total, then there is (x, y) such that $\mathrm{Tr}\left(Q \cup \{(x, y)\}\right) \supsetneq Q$, contradicting the maximality of Q.